LESSON 11 · CHEAT SHEET

# Charts & Excel Output

## EVERY CHART SCRIPT STARTS LIKE THIS

```
import matplotlib
matplotlib.use("Agg")        # no window, files only
import matplotlib.pyplot as plt
import pandas as pd
from pathlib import Path
Path("my_work/output").mkdir(parents=True, exist_ok=Tru
df = pd.read_csv("data/thornmere_payroll.csv")
```

## THE CHART RECIPE (ALWAYS THIS ORDER)

```
plt.figure(figsize=(8, 4))   # 1 new picture (inches)
plt.bar(x, y)                # 2 draw
plt.title("..."); plt.xlabel("..."); plt.ylabel("...")
plt.xticks(rotation=30, ha="right")
plt.legend()                 # if you used label=
plt.tight_layout()           # 3 stop labels being cut
plt.savefig("my_work/output/x.png", dpi=150)  # 4
```

## CHART TYPES

| | |
|---|---|
| `plt.bar(labels, values)` | vertical bars |
| `plt.barh(labels, values)` | horizontal bars |
| `plt.hist(values, bins=12)` | histogram |
| `plt.plot(x, y)` | line |
| `plt.axvline(v, linestyle="--")` | vertical marker line |
| `color="#1f3a5f"`, `label="Men"` | per series |

## GROUPED BARS (TWO SERIES)

```
x = np.arange(len(table.index))
w = 0.4
plt.bar(x - w/2, table["F"], w, label="Women")
plt.bar(x + w/2, table["M"], w, label="Men")
plt.xticks(x, table.index, rotation=30, ha="right")
```

## PANDAS → EXCEL

```
df.to_excel("out.xlsx", sheet_name="Data")
df.to_excel("out.xlsx", index=False)   # no row numbers

with pd.ExcelWriter("out.xlsx") as writer:
    t1.to_excel(writer, sheet_name="By dept")
    t2.to_excel(writer, sheet_name="Top 10", index=Fals
```

Needs `openpyxl` installed. Sheet names: max 31 chars, no `/ \ ? * [ ]`.

## OPENPYXL: MAKE IT PRETTY

```
from openpyxl import load_workbook
from openpyxl.styles import Font, PatternFill
wb = load_workbook("out.xlsx")
ws = wb["Data"]                  # or wb.active
for cell in ws[1]:               # header row
    cell.font = Font(bold=True, color="FFFFFF")
    cell.fill = PatternFill("solid", fgColor="1F3A5F")
for row in ws.iter_rows(min_row=2, min_col=3, max_col=5
    for cell in row:
        cell.number_format = '#,##0.00 "cr"'
ws.column_dimensions["A"].width = 16
ws.freeze_panes = "A2"
ws2 = wb.create_sheet("Chart")
ws2.add_image(Image("chart.png"), "B2")  # openpyxl.dra
wb.save("out.xlsx")
```

## USEFUL CELL FACTS

| | |
|---|---|
| `ws["C2"].value` | what's in C2 |
| `ws.max_row`, `ws.max_column` | size of the sheet |
| `wb.sheetnames` | list of sheet names |
| `"0.0%"`, `"#,##0"`, `"dd/mm/yyyy"` | other number formats |

### GOTCHAS

- No `plt.show()` in scripts. Save, then open the file.
- Forgot `plt.figure()`? Charts pile up on one picture.
- Labels cut off? You forgot `tight_layout()`.
- matplotlib colour `"#1F3A5F"`, openpyxl colour `"1F3A5F"`.
- Workbook open in Excel = "Permission denied" when saving. Close it.